Data Platform Engineer — Take-Home Exercise

KT Grant · Finance · October 2026

Why this exercise

This role turns ERP exports into numbers Finance can trust, and then builds the platform that delivers them. The exercise is a small version of that job. It uses **Northline Industrial Services**, a fictional company with synthetic data, so nothing here is confidential.

We care about how you approach the problem, not which tools you use. **AI tools are allowed and encouraged.** Use whatever language, platform or assistant you like. We only ask that you tell us how you used them (Part 5).

Time and delivery

- We expect about **4 to 6 hours** of work. Please return it within **3 business days** of receiving it.
- Deliver in any format: a ZIP, a Git repository, a notebook link (Colab, Fabric, Databricks) or a PDF with code attached.
- One file is required in a fixed format, so we can check results the same way for every candidate (Part 1).
- After you submit, we will schedule a **30-minute walkthrough** to review your work together.

The situation

Northline's Finance team reports quarterly **repair and maintenance (R&M)** cost by branch. The data comes from two ERPs: ERP-A for the four US branches (a transaction export) and ERP-B for the Canadian branch (a GL extract in CAD). Accounting closed Q3 and reports total R&M of **$14,120,360.57**. Finance wants its own number, built from the source data, and wants to know whether Accounting's figure is right.

The attached data pack holds the files and a **README with the Finance rules**. Read it first. The files come as they would from the systems: they have not been cleaned.

Your tasks

Part 1 — Build Q3 R&M (required output). Build a reproducible process that turns the raw files into Q3 2026 R&M in USD by branch and month. Submit the result as **rm_q3_by_branch.csv** with exactly these columns:

Column	Format	Example
Branch	RIV, OAK, PIN, ELM or LKS	RIV
Month	YYYY-MM	2026-07
RM_USD	Number, 2 decimals, no symbols	1234567.89

Include your code, notebook or workflow so we can rerun it. If you use a GUI tool, include screenshots of every step.

Part 2 — Data quality log. List every data issue you found. For each one: what it was, how you detected it, the rule you applied, and the number of rows and USD affected. A table is fine.

Part 3 — Reconcile to Accounting. Bridge your Q3 total to Accounting's $14,120,360.57, item by item, **to the cent**. For each reconciling item, say who is right, the evidence, and what should change and where.

Part 4 — Design memo (one page maximum). Northline now wants this automated across the whole group:

- 6 legal entities on 4 different ERPs, a few tens of thousands of transaction rows per week in total.

- The main ERP's database **locks tables when queried live** during business hours. IT can restore a copy from the nightly backup.
- The company runs on Microsoft 365, uses Power BI, and wants AI tools to answer questions from the same governed numbers.

Tell us how you would get the data out, which platform you would choose and why (with a rough monthly cost), where the Finance rules from the README would live and who would own them, and what you would deliver in the first 30 days.

Part 5 — How you used AI. A short log, half a page is enough:

- Which tools you used and for what.
- Two or three of your most useful prompts or instructions.
- At least one thing an AI tool got wrong or missed, and how you caught it. If nothing went wrong, explain how you checked.
- Your estimate of how much of the code was AI-written.
- Using AI heavily is fine. Submitting without this log is not.

How we will evaluate

Area	What we look for
Correct numbers	Your Part 1 file matches the Finance rules applied to the data.
Data judgment	You found the issues, chose the right treatment and can prove it.
Reconciliation	Your bridge closes to the cent and the explanations are backed by evidence.
Design	Practical, right-sized for this volume, clear on cost, ownership and first steps.
AI use and ownership	You use tools well, verify what they produce, and can explain and change your own work live.

If something in the data or the rules is ambiguous, make a reasonable decision, write it down, and keep going. Questions can go to the recruiter, but the README has what you need.